ROBOTICS CONTROL UNIT

PROJECT NAME:	ROBOTICS CONTROL UNIT
PROJECT GROUP:	SAHIL KHANDARE, PRATIK KHADSE & VEDANTI DAMDHAR

WHY THIS PROJECT ?

- This project was undertaken to develop a custom ROBOTICS CONTROL UNIT for driving motors according to specific system requirements. The objective is to design and build an in-house motor driver that can be tailored and optimized as performance needs evolve in future competitions undertaken by the club.

INTRODUCTION

- The project was to be used to replace the L298 series Motor Driver being used at our club.
- The whole Design and Development was done for Line follower and RC Car.

KEY FEATURES

1. Dual Motor output capable of 2 amps of current.
2. Cortex M0+ MCU @ 64MHz with 32kB RAM and 128kB FLASH which can be used for data logging as well.
3. Multiple input for sensors, 1 UART and 1 I2C Channel.

ARCHITECTURE

1. Microcontroller Integration:
   STM32G070KBT6
2. DRIVER Selection:
   DRV8837 BY TEXAS INSTUMENTS

RESULTS

- The development of the motor driver for the Robotics Control Unit demonstrated the feasibility of in-house motor control electronics for robotic applications. The project resulted in a functional design approach for driving motors with controlled speed and direction, while emphasizing the importance of protection circuitry, power handling, and signal integrity. Through this process, practical insights were gained into motor control principles, hardware reliability, and system-level integration, which will directly inform improved and more robust motor driver designs in future iterations.

FUTURE SCOPE

1. ☐ **Closed Loop Control**
2. ☐ **ESD Protection**
3. ☐ **PID Tuning and Field Oriented Control for smoother control**